

Nima Nilchian

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Education

Kharazmi University

B.Sc. in Computer Engineering

Tehran, Iran

Sep 2020 – Jan 2025

- Total GPA: 17.72/20 or 3.78/4 (Credits:139)
- Ranked top 10% among fellow Computer Engineering Students

Research Interests

- Natural Language Processing
- Large Language Models
- Information Retrieval
- Deep Learning
- Applied Machine Learning
- Knowledge Graphs

Publications

O.K. Gargari, G. Habibi, **N. Nilchian**, A. Shafiee. "Comparative Analysis of Large Language Models in Psychiatry and Mental Health: Focus on GPT, AYA, and Nemotron-3-8B", Asian Journal of Psychiatry, 2024.

Research Experiences

Farzan Clinical Research Institute

Apr 2024 – Present

Evaluating LLM's in Psychiatry and Mental Health,

- Involved in multiple research projects **evaluating LLMs in psychiatry and medical domains.**
- Researching on **prompt engineering** techniques to **optimize model output** for desired applications.

Remote Research Assistant at University of Calgary

Jun 2023 – Sep 2023

UWB Real-time Locating Systems (RTLS), Supervisors: [Dr. Askarian](#) , [Dr. Sadeghpour](#)

- Increasing the accuracy of Auto-Positioning with UWB-RTLS devices, for Automation in Construction.
- Worked on **Object Detection, Dynamic Tracking and Position Estimation - Data Analysis, Visualization, Optimization - Embedded Programming (C), Raspberry Pi.**

Work Experiences

AI Software Engineer

[Smilinnno Innovation hub](#), Iran

Jan 2025 – Present

- Developed and optimized an agentic RAG application, focusing on performance enhancements.
- Improved system performance by implementing asynchronous processing and enhancing overall scalability.
- Responsible for backend software development and integrating AI features.

NLP Engineer

[Farzan Institute](#), Iran

Oct 2023 – Jan 2025

- **Continual-Learning** and **Fine-tuning** LLMs for mental health use cases.
- Developed a **Chatbot** that acts as a therapist. It engages in multi-turn conversations, asks follow-up questions, and generates a comprehensive summary and recommendations at the end.
- **RAG system** with mental health sourcebooks for domain-specific questions.
- **Symptom Checker** which identifies the relevant medical questionnaire based on user symptoms.

Machine Learning Engineer Intern

[Roshan](#), Tehran, Iran

July 2023 – Aug 2023

- **Key-Phrase Extraction:** Utilizing statistical and Graph-Based algorithms, transition toward Embed-Rank and Sentence Transformers for Persian text as a low-resource language.

President of Student Scientific Association of Computer Engineering

Kharazmi University,

March 2023 – May 2024

- Responsible for holding various events, workshops and meetings.

Editor-in-Chief of Pointer Magazine

Kharazmi University,

Oct 2022 – Sep 2023

- Author, reviewer, and organizer of [Pointer Magazine Issue 19](#). ([outstanding journal](#) in Titr Festival)

Teaching Assistant

Kharazmi University,

- Artificial Intelligence, Fall 2023.
- Advanced Programming, Spring 2024.
- Fundamentals of Programming, Fall 2021, Fall 2023, and Fall 2024.

Selected Projects

Automated IELTS Essay Evaluation Platform

Fall 2024

Bachelor's software project

- Automated IELTS Essay Scoring (AIES) [project demo available here](#)
- Evaluated multiple LLMs performance on IELTS essays with QWK and MAE metrics, with visualizations.
- developed feedback generation platform utilizing: LLMs with Langgraph, Fastapi backend, React.js frontend with deployments.

Retrieval Augmented Generation (RAG) in Mental-Health Domain

Spring 2024

Farzan Institute

- **High-Quality** RAG chatbot, rigorous **evaluations**, **Smart-Chunking** concept, **Prompt Engineering**.

Integrated Evaluation Platform for LLMs

Spring 2024

Farzan Institute

- **Evaluating** the efficiency and improvements of *Fine-tuned* or *RAG* models outcomes with multiple-choice or open-ended questions with various lexical, similarity, numerical analysis reported in a structured format.

Binary Stress Detection

Summer 2023

IPM Summer Bootcamp

[Github](#)

- **Fine-tuned**, evaluated and compared **LLMs** for binary stress detection and stress category classification.

Web Information Retrieval (3)

Spring 2023

Modern Information Retrieval – Prof. Mahdiah Soleymani (Sharif University of Technology)

[Notebook](#)

- **Crawled** 2000 papers. **Link Analysis** using **PageRank** and **HITS** algorithms. Implemented **Personalized PageRank**, **Personalized Search Engine** and **Recommender System**.

Machine Learning and Deep Learning Approach To Information Retrieval (2)

Spring 2023

Modern Information Retrieval – Prof. Mahdiah Soleymani (Sharif University of Technology)

[Notebook](#)

- Implemented and Evaluated **Naïve Bayes** Classifier (from scratch) - **Neural Network Model** and **Transformer Based** Classifier to classify documents labels.
- **Clustering** documents (**K-means** and **Hierarchical Clustering**), **visualizing** and **evaluating**.

Traditional Information Retrieval (1)

Winter 2023

Modern Information Retrieval – Prof. Mahdiah Soleymani (Sharif University of Technology)

[Notebook](#)

- **Positional Inverted Index**, **Index Compression**, **Scoring Documents** and **Search system**

Username Generator

Summer 2022

Hamrah Aval Summer AI Bootcamp

[Notebook](#)

- **Encoder-Decoder Sequence Models** for translating English usernames to equivalent Persian names.

Machine Learning Projects

Spring 2023

Quera ML Bootcamp

[Github](#)

- **Search analysis - Intelligent Search System - Polarity Detection**, etc.

Karamoozesh Backend

Fall 2022

Software Engineering and Testing – **First Project** among all by Prof. Mirtaheri.

[Github](#)

- **Project Manager** of a 9 member team in both Software Engineering and Testing courses. **Developed Back-end with Django-Rest Framework**. performed **Unit-Tests** and **Integration-Tests**.

Selected Courses

- Modern Information Retrieval: 19.7/20
- Signals and Systems: 20/20
- Research and Presentation Skills: 19.9/20
- Software Engineering: 20/20
- Artificial intelligence: 17.3
- Linear Algebra: 17.75/20
- Algorithm Design: 18/20
- Software Testing: 19.85/20

Selected Certifications

- **Artificial Intelligence Summer Camp** | IPM Summer 2023
Deep Learning, Computer Vision, NLP, Tensorflow, Pytorch. (ranked 3 out of 140)
- **AI and Python Summer Bootcamp** | Hamrah Aval Summer 2022
Python, Pytorch, Machine Learning, Deep Learning, NLP, Huggingface.
- **Professional Project-Based Course in Machine Learning** | Quera Spring 2023
Python, Scikit-Learn, Machine Learning, Supervised and Unsupervised Learning, Neural Networks.
- **Scrum Foundation Course** | Scrum Alliance Fall 2024
Scrum/Agile, Project Management, Leadership, Communication.
- **Advanced Algorithm Thinking and Data Structure** | Quera Summer 2021
Solved more than 100 advanced algorithm training examples with C++ and Python.

Technical Skills and Interests

- **Open Libraries:** Pytorch, Scikit-learn, OpenCV, Pandas, Matplotlib.
- **Frameworks:** Django, FastAPI, Huggingface, Langchain, Langgraph.
- **Soft Skills:** Project Management, Responsibility, Fast-Learner, Flexibility, Hard Working.
- **Languages:** Persian(Native), English(Advance), Turkish(Basic).
- **Interests & Hobbies:** Playing Music (Hangdrum & Setar) - Yoga - Running - Mountain Climbing.

Volunteer Jobs

Social Media Manager & Translator Of Conscious Planet Movement for Persian Language

This Movement was initiated by Sadhguru for a 100-day ride journey to raise awareness about Soil Extinction.

References

- Jamshid Shanbehzadeh, Full Professor, Kharazmi University, Tehran, Iran; jamshid@khu.ac.ir
- Azadeh Mansouri, Associate Professor, Kharazmi University, Tehran, Iran; a.mansouri@khu.ac.ir
- Leili Mirtaheri, Assistant Professor, University of Calabria, Italy; mirtaheri@khu.ac.ir
- Iman Askarian, Assistant Professor, Sharif University, Tehran, Iran; iman.askarian@sharif.edu